# Stabilizers in geometric halving

Human Compression project

We give a stabilizer proof of the coset identification in Theorems B and 4.4 of [1]. The proof does not require the halved geometry to be residually connected. This distinction matters: the assertion about residual connectedness in the inspected preprint has counterexamples, whereas the coset identification below remains valid. No claim is made about an uninspected later edition.

## 1 The construction and the retained geometric facts

Let $\Gamma$ be a regular hypertope of finite rank $n \geq 3$ with distinguished generators

$$G = \mathrm{Aut}(\Gamma) = \langle a, b, r_2, \ldots, r_{n-1} \rangle, \qquad [a, r_i] = 1 \quad (i \geq 2).$$

The generators are involutions and change, respectively, the objects of types $0, 1, 2, \ldots, n-1$ in the base chamber $(v, \{v, w\}, x_2, \ldots, x_{n-1})$, where $w = av$. Assume that the vertex–edge truncation is a simple graph $Q$, and that an edge is incident with a higher object exactly when both its endpoints are incident with that object. These are conditions (B1) and (B2) of [1].

For a graph $S$, its bipartite double cover $D(S)$ has vertices $(u, \epsilon)$, with $\epsilon \in \{0, 1\}$, and edges $(u, 0)(z, 1)$ whenever $uz$ is an edge of $S$. Choose the component $C$ of $D(Q)$ containing $(v, 0)$. For a higher object $x$, let $Q_x$ be its vertex–edge shadow. The objects of the halved incidence system $L$ are the vertices of $C$, with their two types, and the pairs $(x, C_x)$, where $C_x$ is a component of $D(Q_x)$ contained in $C$. Vertex incidence is graph adjacency or membership in $C_x$; two higher objects are incident when their original objects are incident and their chosen components intersect.

We retain the previously proved component-intersection lemma for this construction. It uses connectedness of the original flag shadows, (B1), and (B2), and does not assume residual connectedness of $L$. Its consequences needed here are that $L$ is a thin geometry and that its chambers correspond bijectively to the original chambers whose oriented edge has its lift in $C$. The component of each higher object is uniquely determined by the lifted first endpoint. In particular, when $Q$ is nonbipartite every original chamber occurs; when $Q$ is bipartite exactly the chambers whose first endpoint belongs to the chosen color class occur. The full retained proof, including extension of every flag to a chamber, is charged as a dependency of this note.

## 2 An elementary parity lemma

**Lemma 1.** *Suppose a group $K = \langle a, b, R \rangle$ acts transitively on the vertices and edges of a connected graph $S$, the stabilizer of a base vertex $u$ is $P = \langle b, R \rangle$, and the edge orbit is that of $\{u, au\}$. Suppose $a, b$ are involutions and $a$ commutes with every member of $R$. Put $K^+ = \langle aba, b, R \rangle$. Then $K^+$ is normal in $K$ with index at most two. It is proper exactly when $S$ is bipartite. In that case it is the subgroup preserving the two color classes. Equivalently, it is the stabilizer of either component of $D(S)$ under the action preserving the second coordinate.*

*Proof.* Conjugation by $a$ exchanges $b$ and $aba$ and fixes $R$. Thus $a$ normalizes $K^+$, and $K = K^+\langle a \rangle$. If $K^+$ is proper, the character $K \longrightarrow K/K^+ \cong C_2$ is constant on cosets of $P$, since $P \leq K^+$. It colors the vertices $gu$ by the coset of $g$. The endpoints $gu$ and $gau$ have different colors, so the coloring is a bipartition of $S$.

Conversely, the bipartition of a connected bipartite graph is unique up to exchange. The action of $K$ on its two classes is a homomorphism to $C_2$. Every element of $P$ fixes $u$ and therefore preserves the classes, while $a$ exchanges them. Consequently its kernel contains $K^+$ and excludes $a$. The index

bound gives equality with the kernel. Finally, an even walk stays in a color class and an odd walk changes it. Hence $D(S)$ has two components in the bipartite case, and one otherwise; their stabilizers have exactly the description asserted. □

## 3 All maximal parabolic stabilizers

Put

$$t_0 = aba, \quad t_1 = b, \quad t_i = r_i \ (i \geq 2), \qquad H = \langle t_0, \ldots, t_{n-1} \rangle, \qquad H_i = \langle t_j : j \neq i \rangle.$$

**Theorem 2.** *The group $H$ acts regularly on the chambers of $L$ and is its full group of type-preserving automorphisms. In its base chamber, the stabilizer of the object of type $i$ is exactly $H_i$. Consequently $L$ is isomorphic to the typed coset incidence geometry of $H$ and $(H_i)_{0 \leq i < n}$. None of these assertions requires residual connectedness of $L$.*

*Proof.* Apply Lemma 1 first to $G$ and $Q$. The original vertex stabilizer is $P = \langle b, r_2, \ldots, r_{n-1} \rangle$; this is the standard parabolic stabilizer in a regular hypertope. If $Q$ is nonbipartite, then $H = G$. If it is bipartite, $H$ has index two and preserves the chosen class. The chamber correspondence above and the regular action of $G$ on original chambers show that $H$ acts regularly on the chambers of $L$.

The stabilizer of $(v, 0)$ is $P = H_0$. The stabilizer of $(w, 1)$ is $aPa = \langle aba, r_2, \ldots, r_{n-1} \rangle = H_1$. Both subgroups are contained in $H$, so these are also the stabilizers for the restricted action.

Fix $i \geq 2$ and set

$$K_i = \langle a, b, r_j : j \geq 2, \ j \neq i \rangle.$$

This is the full stabilizer of $x_i$ in $G$. In the original residue at $x_i$, it is transitive on vertices and edges, and its vertex stabilizer is $\langle b, r_j : j \geq 2, \ j \neq i \rangle$. The shadow graph $Q_{x_i}$ is connected: every truncation on at least two types of a residually connected geometry is connected. Its edges are the translates of the base edge. Thus Lemma 1 applies inside $K_i$, even if its action on the shadow graph has a kernel. It says that the stabilizer in $K_i$ of the chosen component of $D(Q_{x_i})$ is precisely

$$K_i^+ = \langle aba, b, r_j : j \geq 2, \ j \neq i \rangle = H_i.$$

An element fixing $(x_i, C_i)$ must fix $x_i$ and preserve $C_i$. Since $H_i \leq H$, its stabilizer in $H$ is exactly $H_i$.

The elements $t_i$ carry the base chamber to its unique $i$-adjacent chamber. For $i \geq 2$ this follows from the old adjacency and the uniqueness of the chosen components through the retained endpoint. The element $b$ fixes $v$ and changes the other endpoint of the base edge, so it gives adjacency of type 1. Conjugating this operation by $a$ gives $aba$, which fixes $w$ and changes the endpoint of type 0. It preserves every higher component, because it fixes the corresponding higher object and the retained lifted endpoint. These are genuine changes, by thinness of the original panels. Since the $t_i$ generate $H$ and $H$ is chamber-transitive, the chamber graph of $L$ is connected. In a thin geometry, a type-preserving automorphism fixing a chamber fixes each adjacent chamber; induction along galleries shows that it fixes every object. Thus the action of the full automorphism group on chambers is free. The regular subgroup $H$ is already transitive, and hence is the full automorphism group.

Lastly send a left coset $hH_i$ to the image under $h$ of the base object of type $i$. The stabilizer calculation makes this a bijection on each type. Intersecting cosets give incident objects. Conversely, incident objects extend to a chamber of $L$, and chamber transitivity supplies an element in the two corresponding cosets. This proves the incidence isomorphism, with distinct types retained even if subgroups coincide. □

The theorem proves the literal coset assertion of [1]; it does not infer the subgroup intersection property from chamber regularity. A stabilizer of a flag of two or more higher objects need not be generated by the remaining $t_i$. That is precisely where the previously established parity condition and the counterexamples to residual connectedness enter. The present argument supplies a missing proof of a known assertion, not a newly solved open conjecture.

## References

[1] C. A. Piedade and P. Tranchida, *Constructing new geometries: a generalized approach to halving for hypertopes*, arXiv:2405.19050v1 (29 May 2024), Theorems B and 4.4, Proposition 4.3.